GenIE | Flow comparison

Shared phases aligned across four registered Python flows · grouped steps, current implementation

PHASE	ISS · default DefaultGenIEFlow	ISS · extended evaluation ISSEvaluationFlow	Performance simulation PerfSimGenIEFlow	Hardware-aware selection HardwareSelectionFlow
Setup	**Setup & checks** Tools · session environment	**Setup & checks** Tools · session environment	**Setup & checks** Tools · session environment	**Setup & checks** Tools · session environment
Baseline	**Benchmark & instruction trace** MLonMCU.Bench / Trace	**Benchmark & instruction trace** MLonMCU.Bench / Trace	**PerfSim benchmark & trace** PerfSimBench / PerfSimTrace	**Benchmark & instruction trace** MLonMCU.Bench / Trace
Discovery	**Discover candidates** Load → analyze → pick → CDFG → query	**Discover candidates** Load → analyze → pick → CDFG → query	**Discover candidates** Load → analyze → pick → CDFG → query	**Discover candidates** Load → analyze → pick → CDFG → query
Generation	**Generate instructions & core** GenerateInstrs / GenerateETISSCore	**Generate + encoding metrics** GenerateInstrs / GenerateETISSCore AssignEncodingScore	**Generate instructions & core** GenerateInstrs / GenerateETISSCore	**Generate + encoding metrics** GenerateInstrs / GenerateETISSCore AssignEncodingScore
Retargeting	**LLVM + functional ISS** Combined / split LLVM · ETISS	**LLVM + functional ISS** Combined / split LLVM · ETISS Compiler-support annotation	**LLVM + timing simulator** HLS scheduling → PerfSim retargeting RetargetPerfSim replaces RetargetISS	**LLVM + functional ISS** Combined / split LLVM · ETISS Compiler-support annotation
Evaluation	**Benchmark & compare** Aggregate + per-instruction evaluation	**Evaluate & reanalyze** Benchmarks + trace / reanalysis Utilization → score → sort	**PerfSim benchmark & compare** Microarchitecture sweep + CPI ranking Per-instruction evaluation retained	**Evaluate & reanalyze** Benchmarks + trace / reanalysis Utilization → score → sort
Refinement	**Filter → regenerate** initial → filtered Generate instructions & ETISS core	**Filter → regenerate** initial → filtered Generate instructions & ETISS core	**Filter → regenerate** initial → filtered Generate instructions & ETISS core	**Filter → select → reevaluate** filtered: ISS evaluation + HLS + scores filtered_selected → final final: ISS evaluation + HLS + synthesis
Outputs	**Report & finalize** Permissions · summary · bundle · cleanup	**Report & finalize** Permissions · summary · bundle · cleanup	**Report & finalize** Permissions · summary · bundle · cleanup	**Report & finalize** Permissions · summary · bundle · cleanup

Reading the figure

Gray: shared phase · green: added evaluation · blue: replacement · amber: expanded hardware refinement
Colors describe structure, not validation. Physical HLS / timing-enabled PerfSim / synthesis validation is pending.
Each column is a sequential flow; refinement expands the evaluation block at explicit candidate stages.